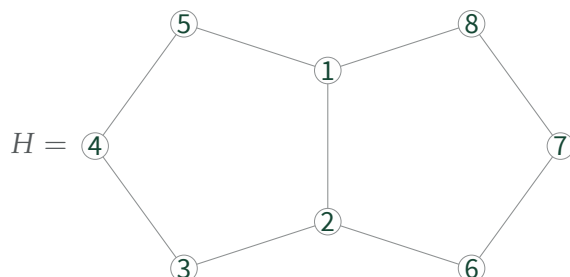


Math 2001 Algebra Exercises

1. Consider the graph H shown below:



One element in the automorphism group of H is the function $f_1 : \{1, \dots, 8\} \rightarrow \{1, \dots, 8\}$ defined by

n	1	2	3	4	5	6	7	8
$f_1(n)$	1	2	6	7	8	3	4	5

List all other elements in the automorphism group of H .

2. Give an example of a graph with 6 vertices with automorphism group equal to the set containing these two functions and only these two functions from $\{1, \dots, 6\}$ to $\{1, \dots, 6\}$ as described below:

n	1	2	3	4	5	6
$f_1(n)$	1	2	3	4	5	6
$f_2(n)$	2	1	4	3	6	5